UNIT 2 LIFE PROCESSES
2.4 Photosynthesis
Markscheme

Part 1

Each question is **[1 mark]**.

1 D

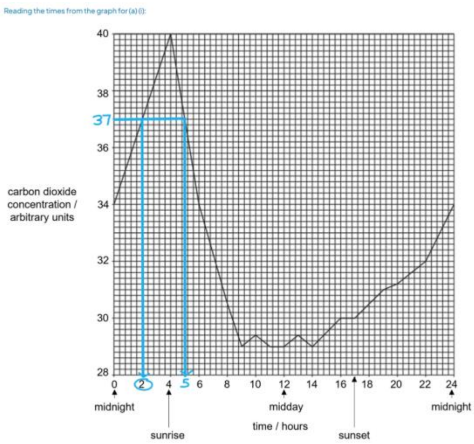
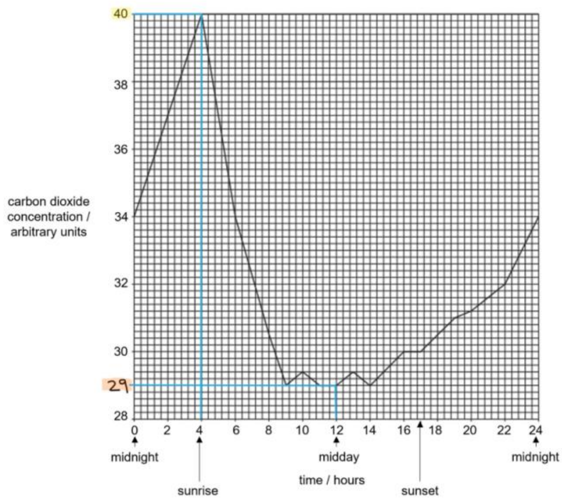
2 C

3 B

4 D

Part 2

5

(a)		Light reactions, in the thylakoid membranes the Calvin cycle (or the dark reactions), in the stroma	[4]
(b)	(i)	02:00 and 05:00  	[1]
(b)	(ii)	40 - 29 = 11 (arbitrary units)  • CO_2 concentration at 04:00 = 40 au • CO_2 concentration at 12:00 = 29 au	[2]

(b)	(iii)	• (Sun)light is present / it is sunrise; • Carbon dioxide is absorbed / used (by the plants); • Photosynthesis is using carbon dioxide faster / at a higher rate than respiration can provide / replace it OR (the rate of) photosynthesis is higher/faster than (the rate of) respiration	[2 max]
(c)		• Did not change / remained the same OR constant [must mention, 1 mark]; • (This is because) the rate of photosynthesis was equal to the rate of (aerobic) respiration; • (This means) the amount of carbon dioxide absorbed (by photosynthesis) was the same as the amount of carbon dioxide released (by aerobic respiration); • (This suggests) the light compensation point is reached. *Remember that respiration is occurring in the plant cells during the day and night. During the day, respiration and photosynthesis are occurring simultaneously. Whether there is a net absorption or release of carbon dioxide by the plant depends on the respective rates of respiration and photosynthesis. When the rates are equal, then there will be no overall increase or decrease in the carbon dioxide levels outside the plant.*	[2 max]
(d)		• No photosynthesis occurred during the night; • No carbon dioxide was absorbed by the plant / the plant only released carbon dioxide (by respiration); • (Likely) no wind dispersed the carbon dioxide.	[1 max]